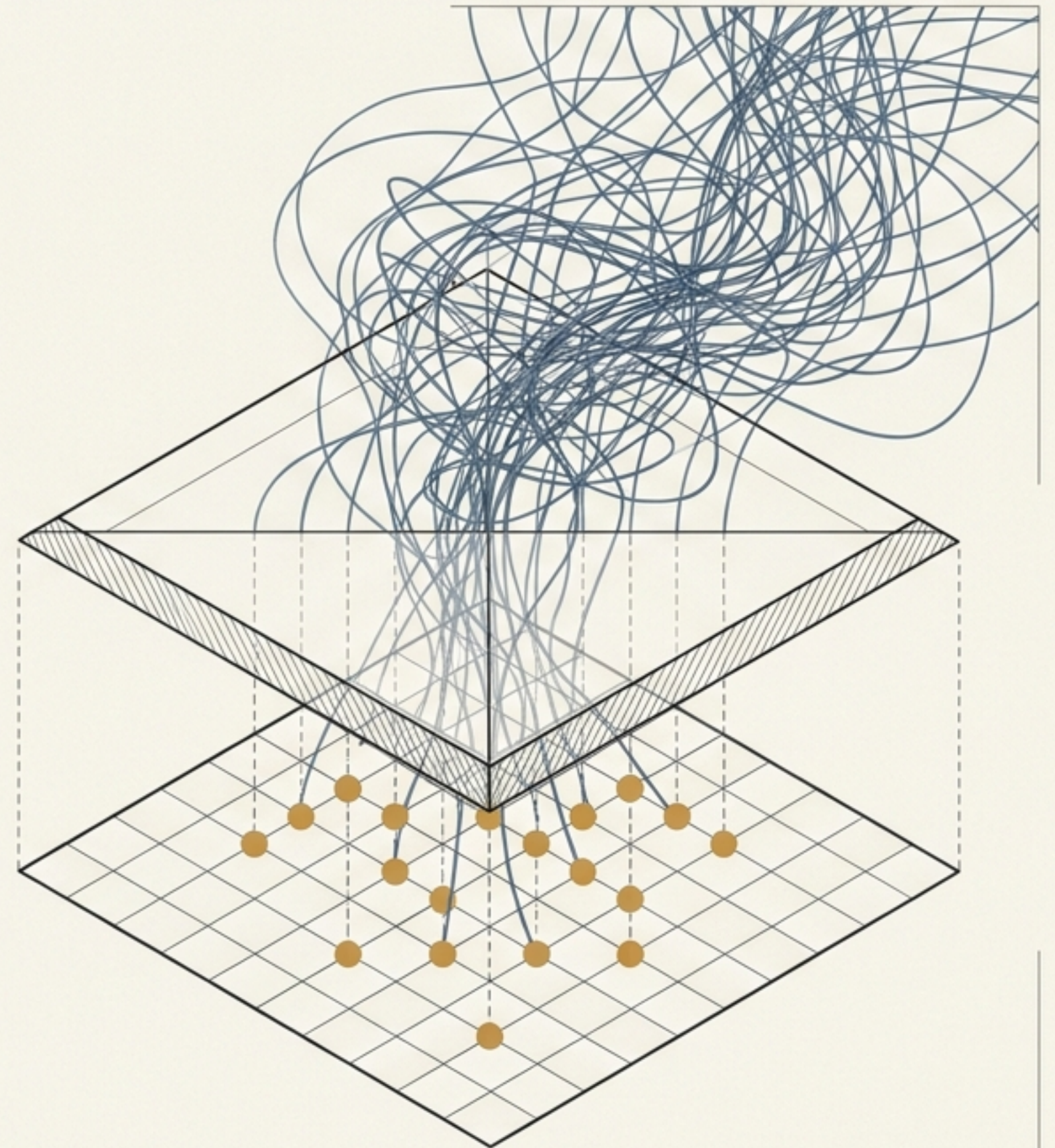


Compressed Causality: The Geometry of Alternative Histories

Stability, Projection,
and the Conditions for
Recovering the Past

Flyxion, 2026



Every system that survives must forget

Every record of the past is an incomplete compression of a history too large to store.
The central question of reconstruction is: under what conditions can a system reconstruct answers about alternative histories from a representation that does not contain them explicitly?

The Mind

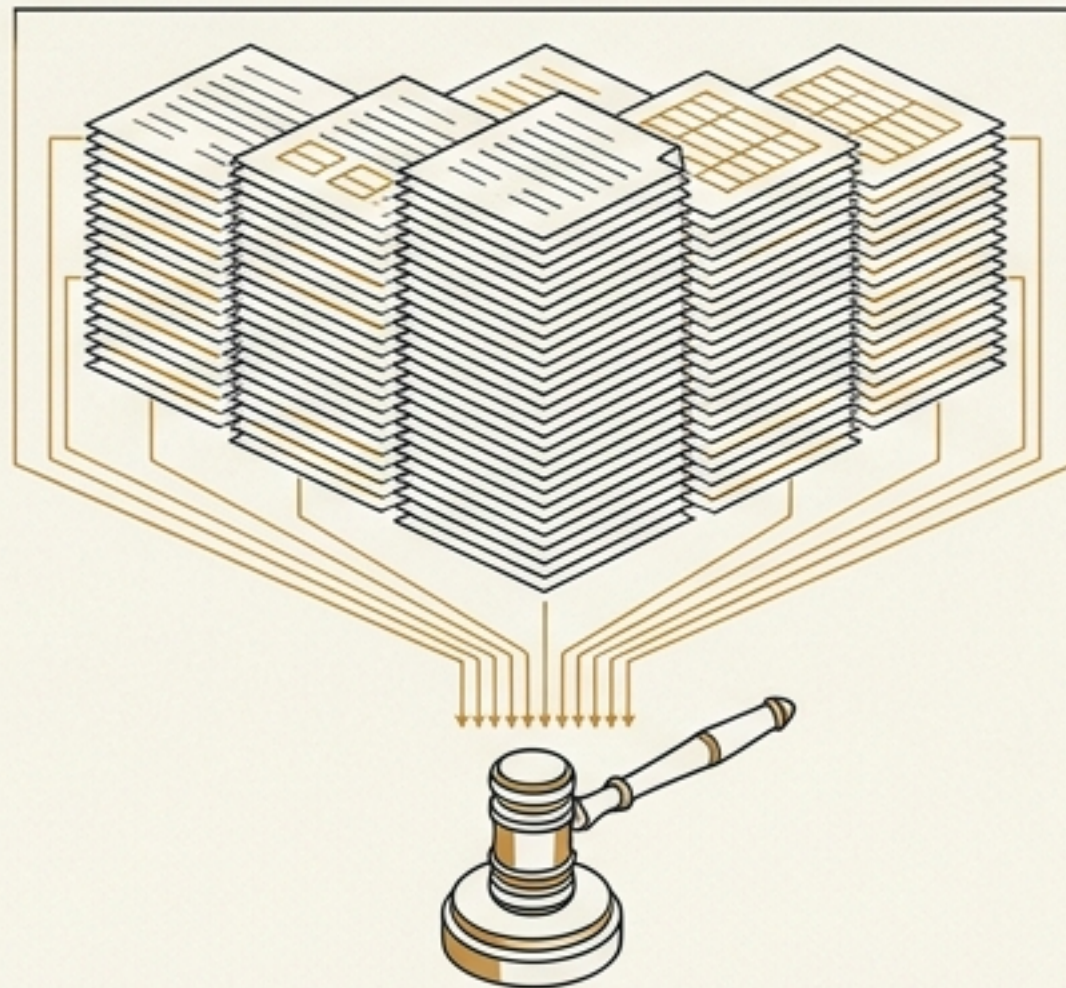
Trajectory Space | Stability



A mind cannot retain every experience; memory is a compression, not a recording.

The Institution

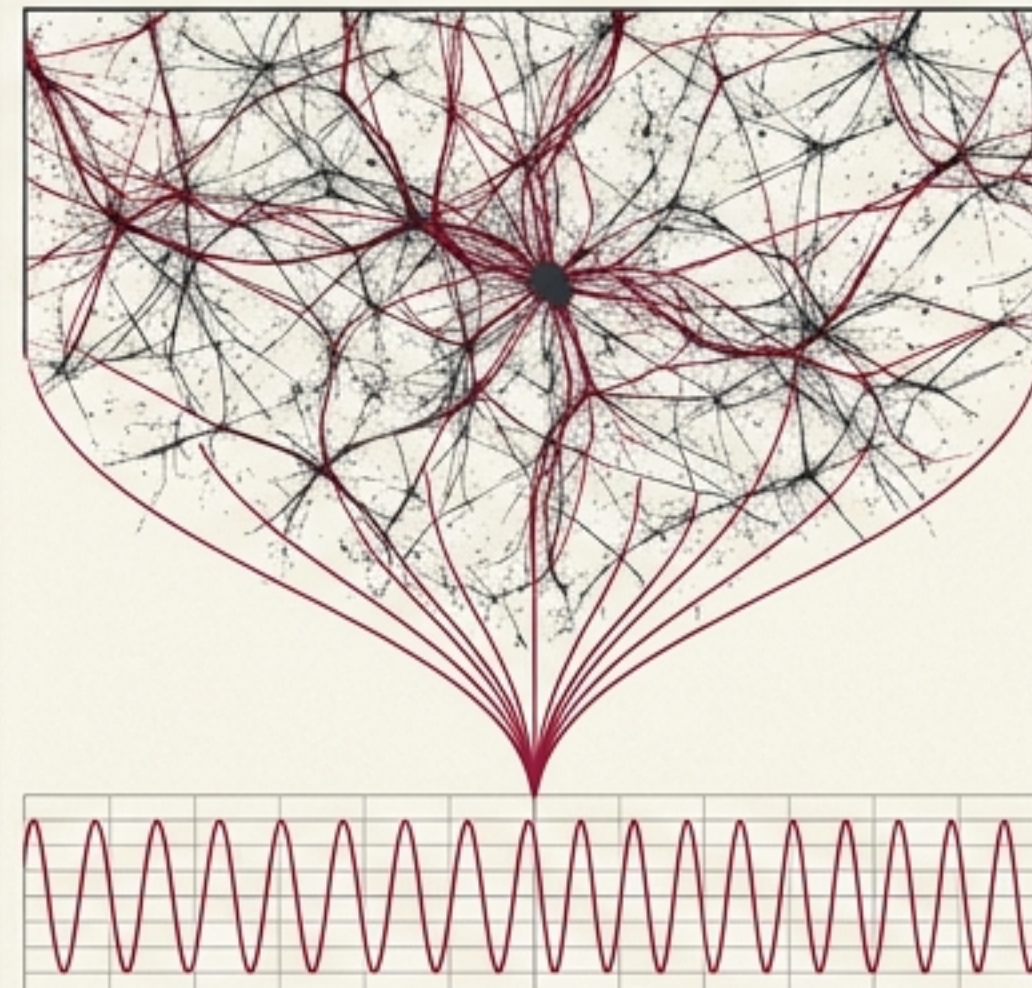
Representation Space | Sufficiency



Institutions preserve only fragments of social reality in precedent and law.

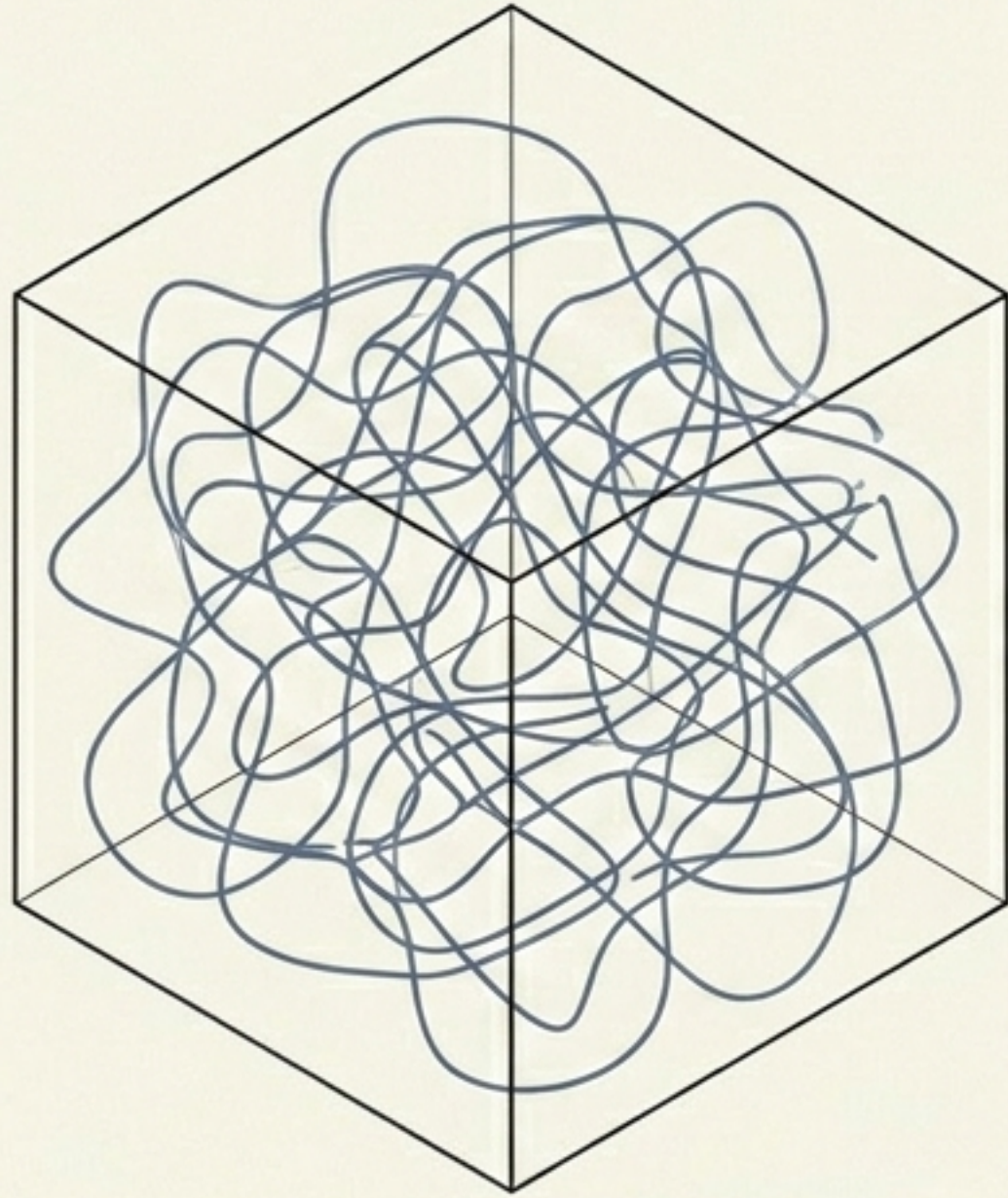
The Universe

Query Space | Reconstruction

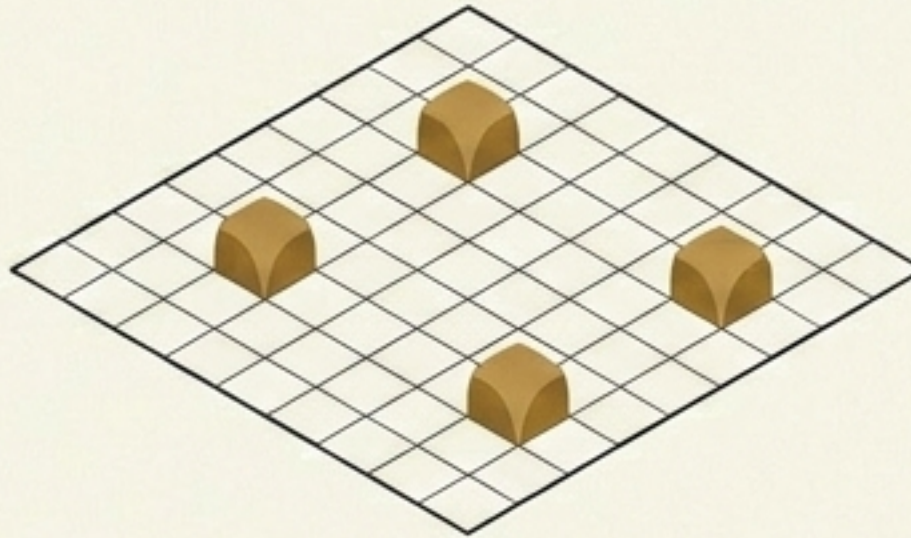


The observable universe contains only a fraction of its own historical information.

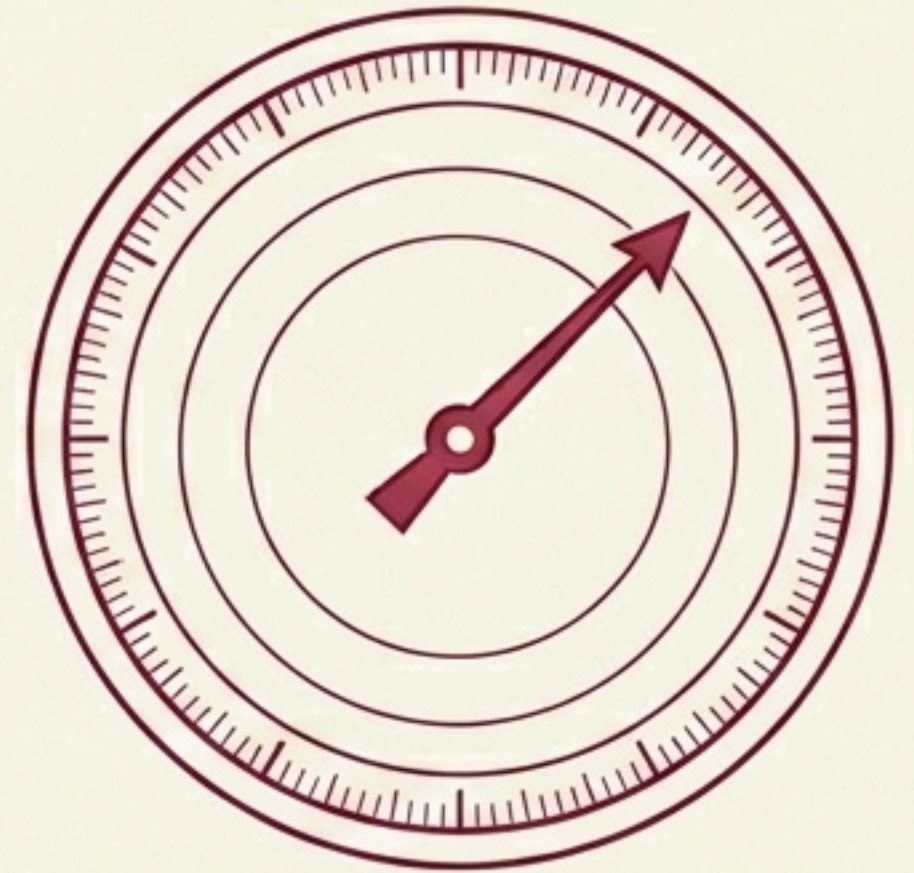
The Three Spaces of History



π



R



Trajectory Space (X)

The ontological space of all possible histories.
A trajectory x answers: what actually happened?

Representation Space (M)

The epistemic space of compressed states.
The remembered world.

Query Space (R)

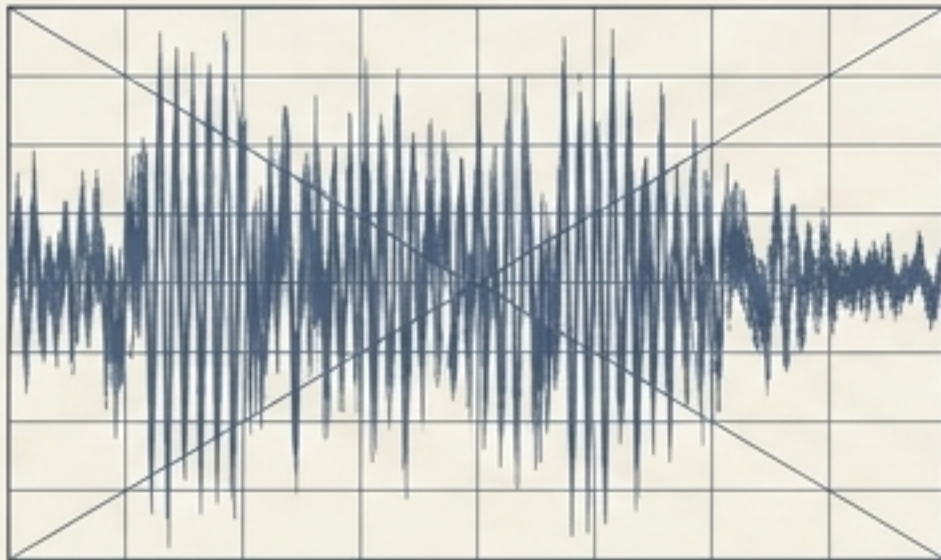
The pragmatic space of answers.
Evaluates the query $Q(x)$ about the past.

The Fundamental Error Decomposition

Every failure of counterfactual reconstruction from a compressed representation can be traced to one or more of three independent sources.

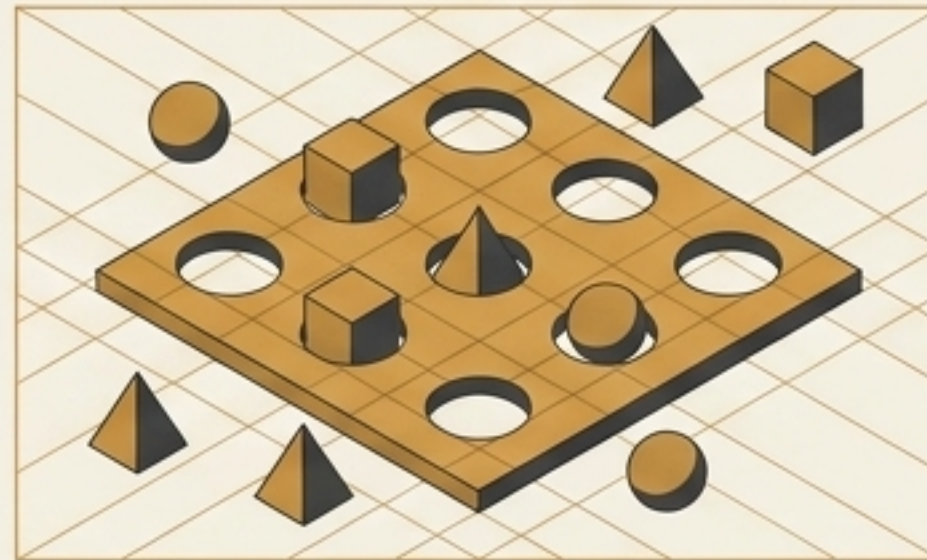
$$E_{total} = E_c + E_s + E_a$$

E_c : Compressibility Error



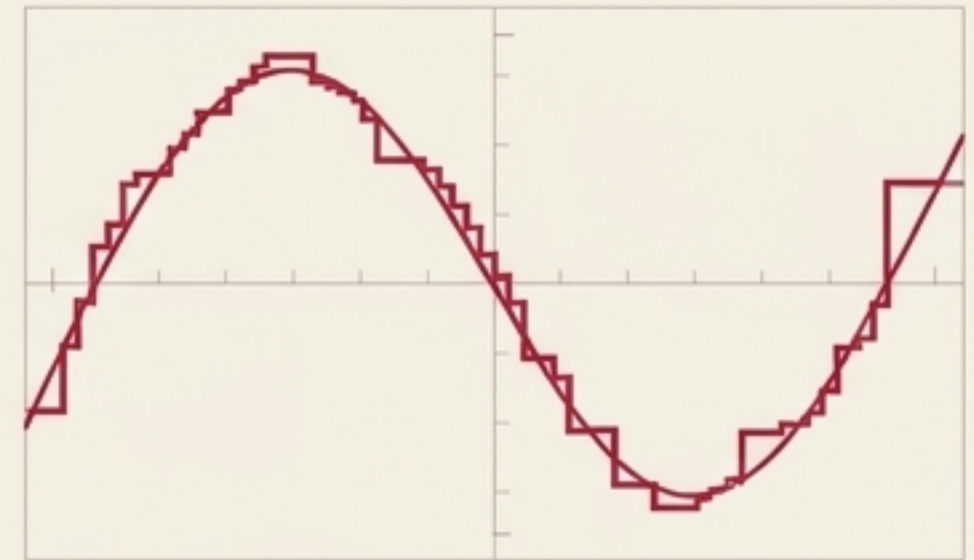
The query's sensitivity to history does not decay with interaction order.

E_s : Sufficiency Error



The compression discards the specific syntactic or structural information the query requires.

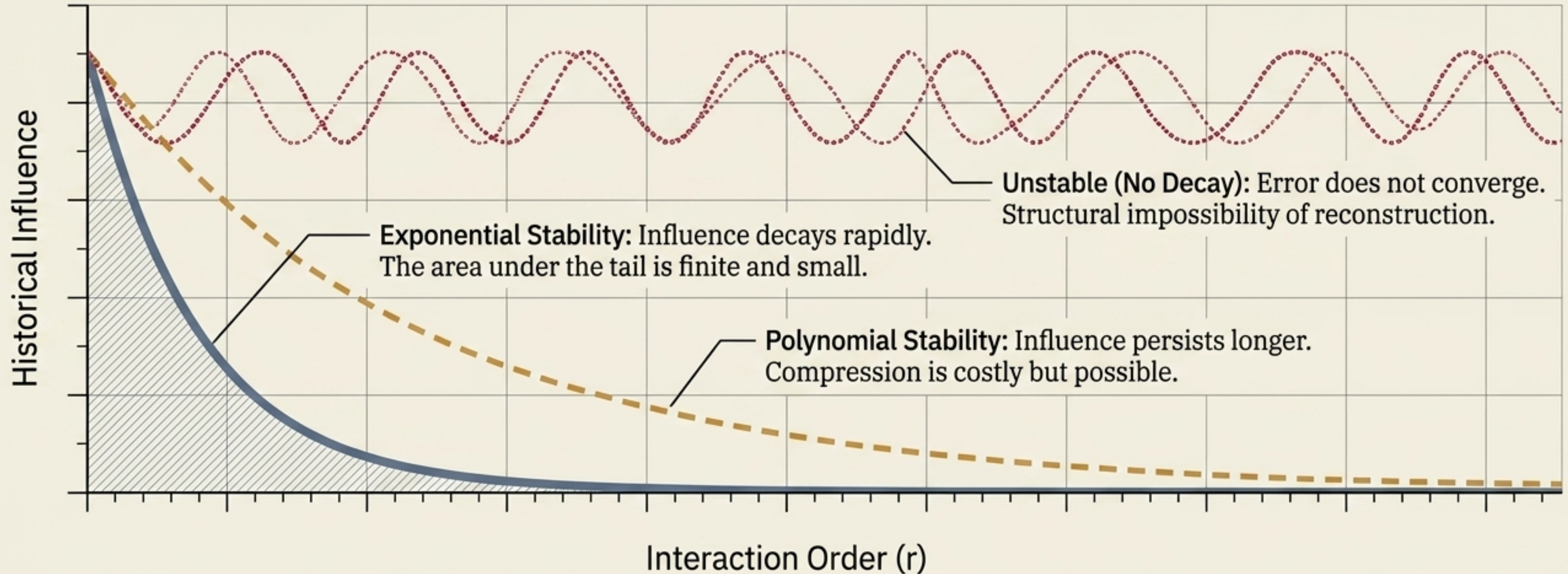
E_a : Approximation Error



The query cannot be represented accurately at the chosen truncation depth, even with perfect stability.

Stability: Why the Past Matters Less

The Interaction Decay Curve

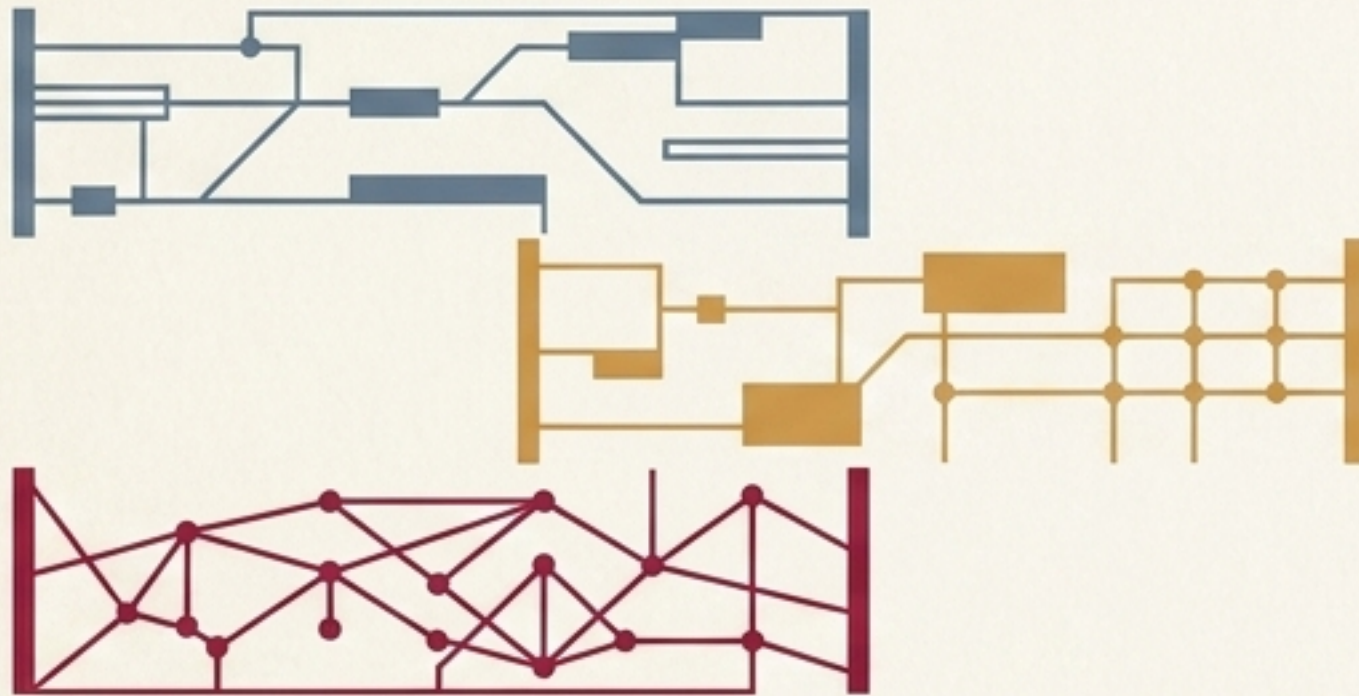


Without stability, no finite compression can capture history's full influence.
Stability is a structural property of the universe, not an algorithm.

Alignment as a Design Principle

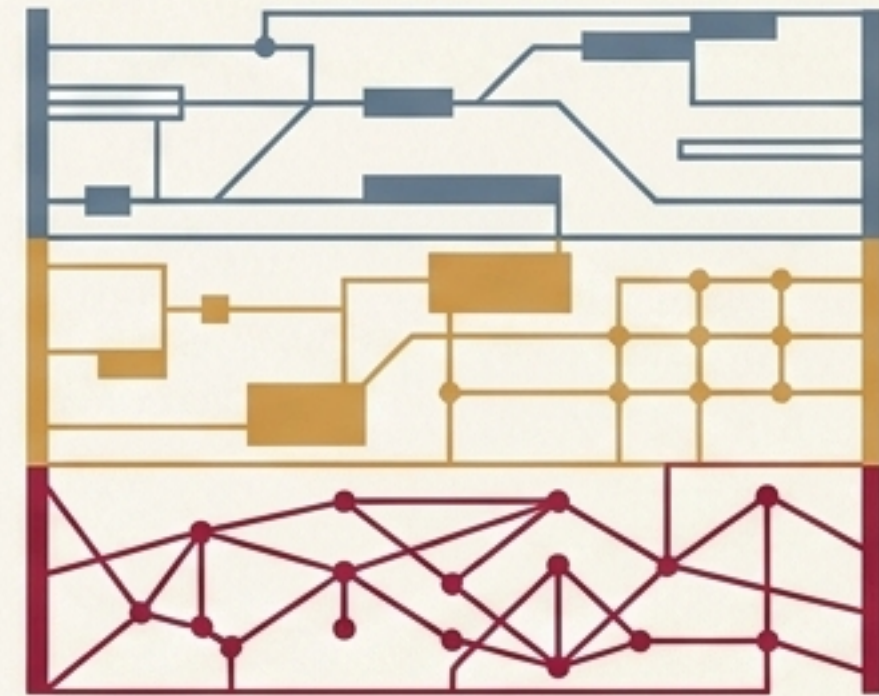
Alignment transfers complexity from query time to design time. An aligned compression forgets exactly what stability says can be forgotten, and preserves exactly what stability says matters.

Unaligned Architecture



Each query pays its own heavy complexity cost. Sufficiency and Approximation errors multiply.

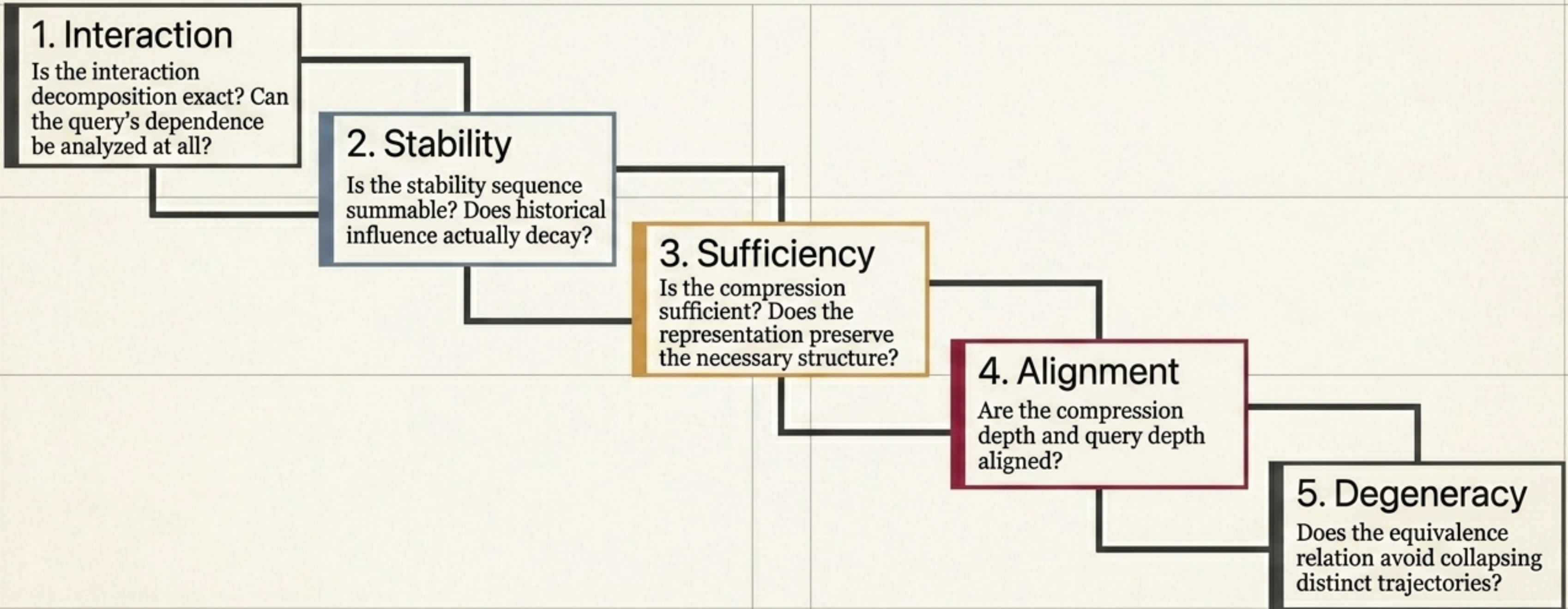
Aligned Architecture



The difficult work is pre-paid during compression design. Errors share a unified decay rate, and sufficiency error is eliminated by construction.

The 5-Question Diagnostic Audit

The Reconstruction Theorem is not just a proof; it is a portable scientific instrument to evaluate any system claiming to reconstruct the past.



The Failure Taxonomy

A reconstruction failure is not the absence of an explanation; it is itself an explanation. The framework distinguishes between limits of design (Architectural) and limits of reality (Structural).

Failure Mode	Core Mechanism	Visual Symptom	Classification	Remedy
Stability Failure	Non-summable decay	Error does not converge	Structural	No remedy at compression level.
Sufficiency Failure	Compression depth < Query depth	Fiber spans multiple query contours	Architectural	Redesign the compression architecture.
Approximation Failure	Suboptimal truncation	High error despite alignment	Architectural	Better approximation family or increased query depth.
Misalignment Failure	Misoriented projection	Preserving the wrong structure	Architectural	Calibrate compression to stability mechanism.
Degeneracy (Type I/II)	Folded quotient space	Distinct histories falsely collapsed	Structural/Arch	Enrich the equivalence relation.

Verified Domains: The Rosetta Stone of Reconstruction

	Machine Learning	Semantic Systems	Physical Fields
Trajectory	Algorithmic Training Histories	Constraint Grammars	Accessibility Dynamics (RSVP)
Interaction Order	Polynomial degree	Branch depth	Transition depth
Stability Mechanism	Derivative decay	Attenuation	Accessibility contraction
Compression	Polynomial sketch	CLIO (branch-tree)	Accessibility skeleton
Alignment Type	By coincidence (computation)	By design (explicit calibration)	Under uncertainty

Applications & Open Empirical Obligations

In unverified domains, the framework serves as a research-program generator, converting vague uncertainty into precise scientific obligations.



Cognition & Memory

Candidate Space: Episodic, Inferential, or Predictive trajectories.

Failure Mode: **False memories**

Open Obligation: Determine the canonical interaction order for human trauma and measure temporal decay.

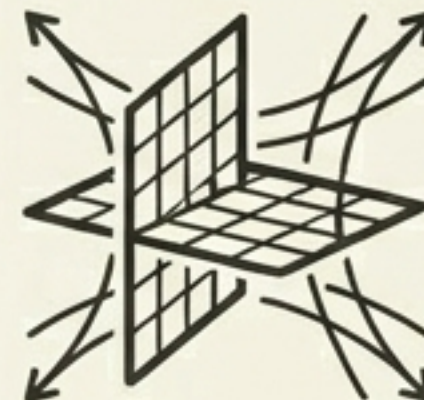


Institutions & Archives

Candidate Space: Legal precedent, administrative records.

Failure Mode: **Historical revisionism**

Open Obligation: Measure institutional influence decay and precedential half-lives.



Cosmology

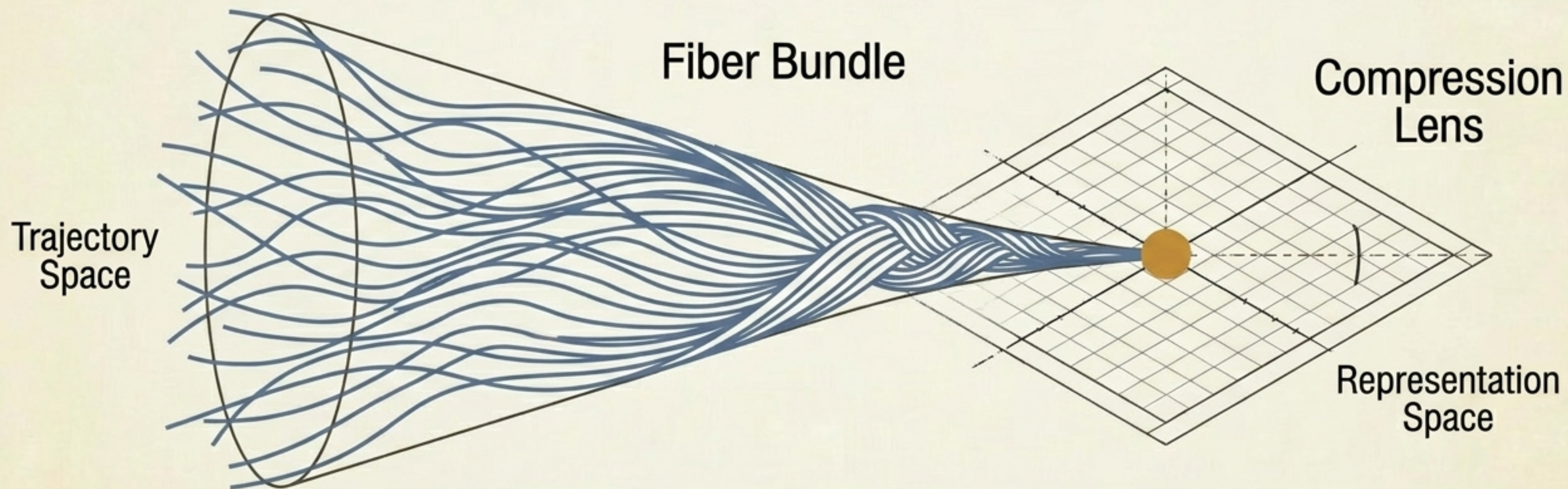
Candidate Space: Spacetime-history, large-scale parameter sequences.

Failure Mode: **Trans-Planckian problem**

Open Obligation: Derive accessibility contraction directly from RSVP field equations.

Alternative Histories as Fibers

Alternative histories are not hypothetical trajectories outside of knowledge. They are trajectories bundled inside the fiber of what current knowledge permits. Counterfactual reasoning is navigation within fibers.



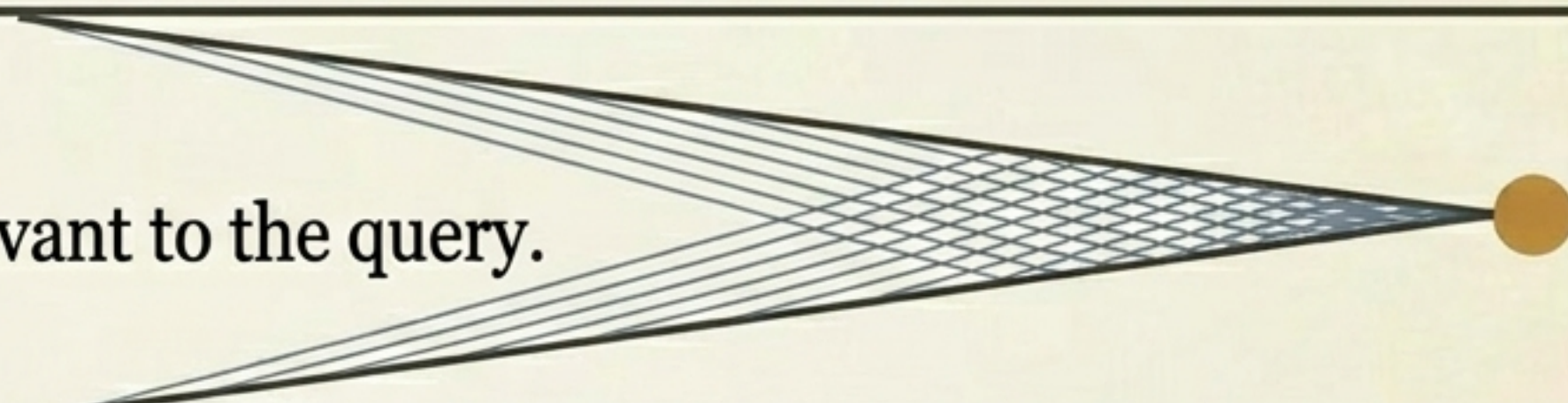
Two trajectories in the same fiber are totally indistinguishable given the compressed representation alone. Stability ensures these fibers are 'query-flat'.

The Cone of Recoverability

The universe dictates how rapidly alternative histories converge. This defines the boundaries of the Cone of Recoverability.

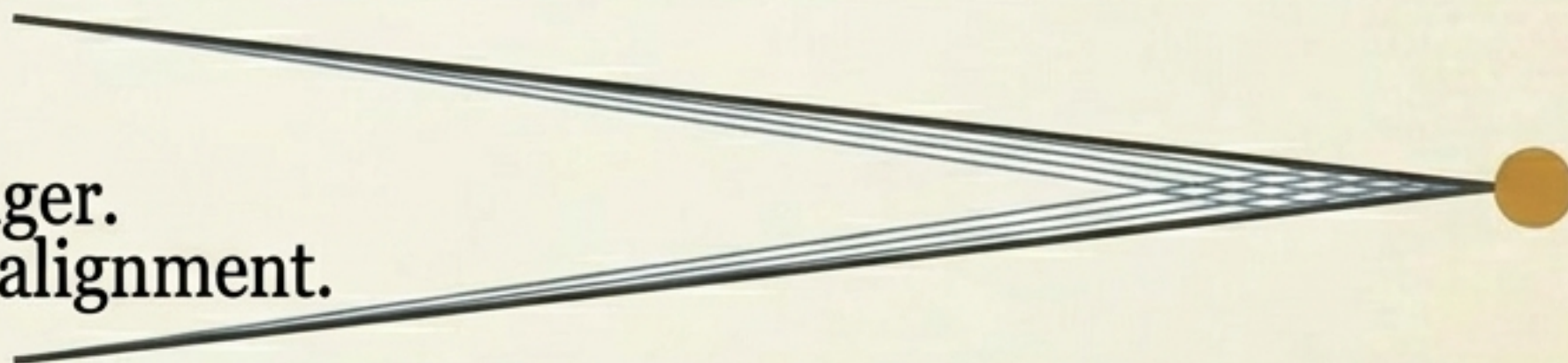
Exponential Stability

Outcome A: History becomes rapidly irrelevant to the query.
Reconstruction is highly tractable.



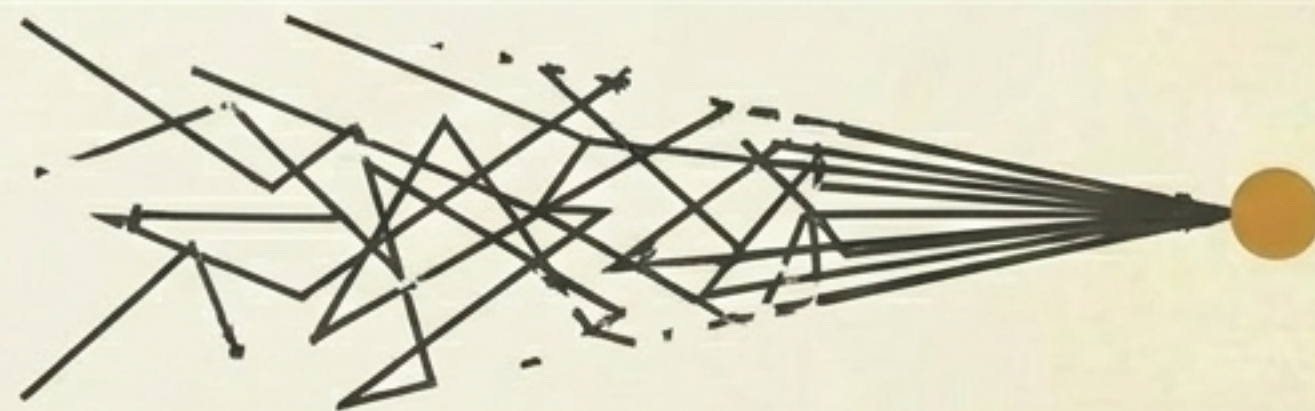
Polynomial Stability

Outcome B: History remains important longer.
Reconstruction is costly and requires deep alignment.



No Stability

Outcome C: History never becomes negligible.
Explanation itself breaks down.

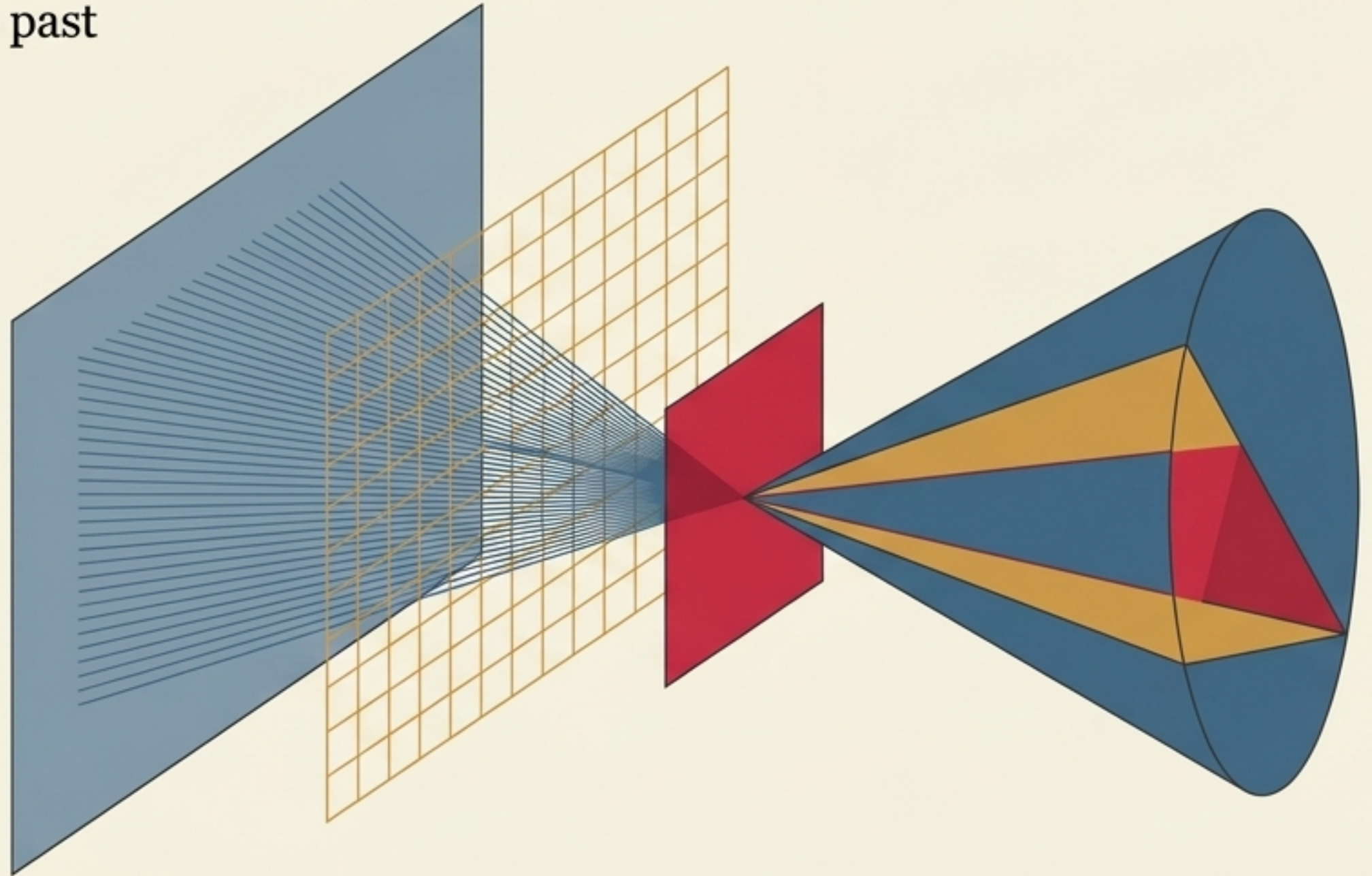


Synthesis: A World That Can Remember

The central question is not whether the past can be reconstructed, but what kind of world makes reconstruction possible.

The Four Geometric Conditions for Memory

1. The trajectory space must be stable (fibers query-flat).
2. The projection must be sufficient (fibers do not cross contours).
3. The projection must be aligned with the stability structure.
4. The equivalence relation must avoid degeneracy.



The geometry of alternative histories is the geometry of recoverable memory itself.